

Biomarker Effect Decomposition

METIMMOX-1 Economic Evaluation

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1 Overview

This report examines how biomarker-positive and biomarker-negative patient subgroups contribute to the overall survival benefit of biomarker-guided treatment strategies. For each biomarker strategy (CRP, TLR, TMB/BRAF), we show:

1. **Survival curves** decomposed into:
 - Biomarker-positive patients (receiving experimental treatment)
 - Biomarker-negative patients (receiving control treatment)
 - Overall strategy (population-weighted average)
 - Control (all patients receiving standard treatment)
2. **Life-years** for each subgroup across different model structures
3. **Model comparisons** showing how results vary by model specification (A, B, C)

1.1 Model Structures and Biomarker Populations

This report uses only the Gamma distribution for all survival models. For detailed model selection and comparison (including Weibull PH models and AIC/BIC statistics), see the Parametric Survival Modeling report (`para_models.qmd`).

Three model structures are compared for each biomarker strategy:

- **Model A (Joint)**: All biomarker-treatment interactions in a single model
- **Model B (Focused)**: All biomarkers as main effects plus one interaction term
- **Model C (Separate)**: Single biomarker with its interaction term only

Table 1 summarizes the patient populations for each biomarker strategy, showing the number of biomarker-positive and biomarker-negative patients and the overall prevalence rate.

Table 1: Biomarker patient populations and prevalence rates.

Biomarker	N Positive	N Negative	N Total	Prevalence (%)
CRP (<5 g/L)	22	43	65	33.8
TLR (>=10%)	41	24	65	64.7
TMB (>=9 mut/MB) or BRAF	29	36	65	44.9

2 CRP Biomarker Strategy

C-reactive protein (CRP) with a cutoff of <5 for biomarker-positive status. **Biomarker prevalence: 33.8%**. Biomarker-positive patients receive alternating FLOX and nivolumab; biomarker-negative patients receive FLOX only.

Table 2: CRP strategy: Overall survival life-years by model structure. Delta columns show difference vs. control.

Model	Control	Biomarker+	Delta+	Biomarker-	Delta-	Strategy	Delta Strategy
Model A	2.17	2.81	0.64	1.73	-0.45	2.09	-0.08
Model B	2.23	2.90	0.68	1.74	-0.48	2.13	-0.09
Model C	2.46	2.81	0.35	1.95	-0.51	2.24	-0.22

Table 3: CRP strategy: Progression-free survival life-years by model structure. Delta columns show difference vs. control.

Model	Control	Biomarker+	Delta+	Biomarker-	Delta-	Strategy	Delta Strategy
Model A	0.90	1.89	0.99	0.84	-0.06	1.20	0.30
Model B	0.95	1.90	0.96	0.86	-0.09	1.21	0.27
Model C	1.07	1.99	0.93	0.99	-0.08	1.33	0.26

2.1 Overall Survival

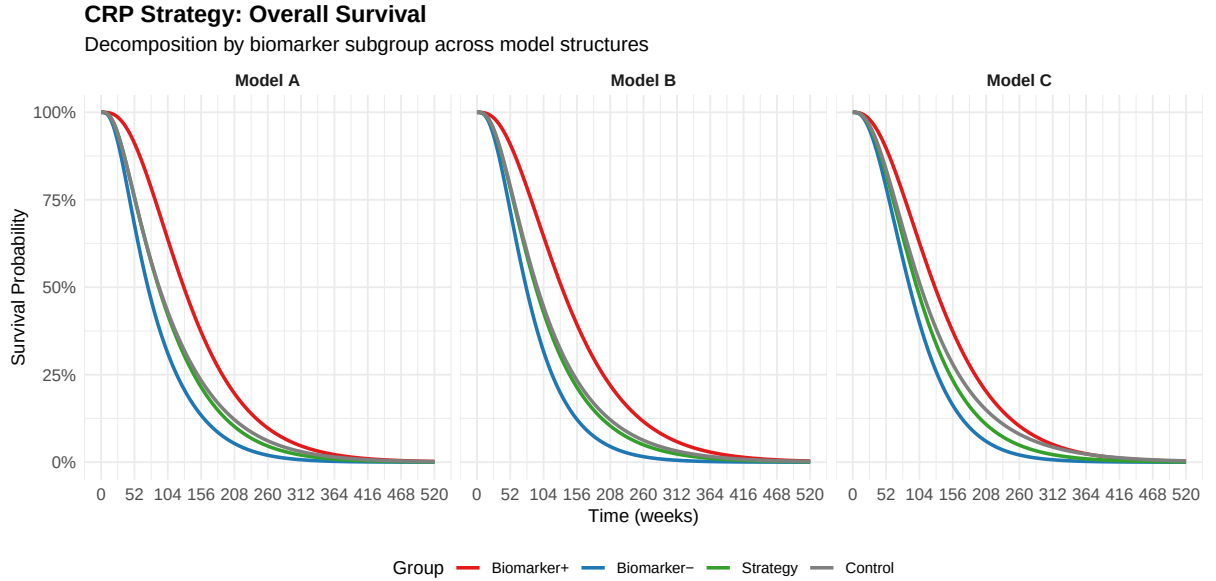


Figure 1: CRP strategy (prevalence: 33.8%) decomposition for overall survival across model structures. Red: biomarker-positive (experimental treatment), Blue: biomarker-negative (control treatment), Green: population-weighted strategy, Gray: control (all patients on standard treatment).

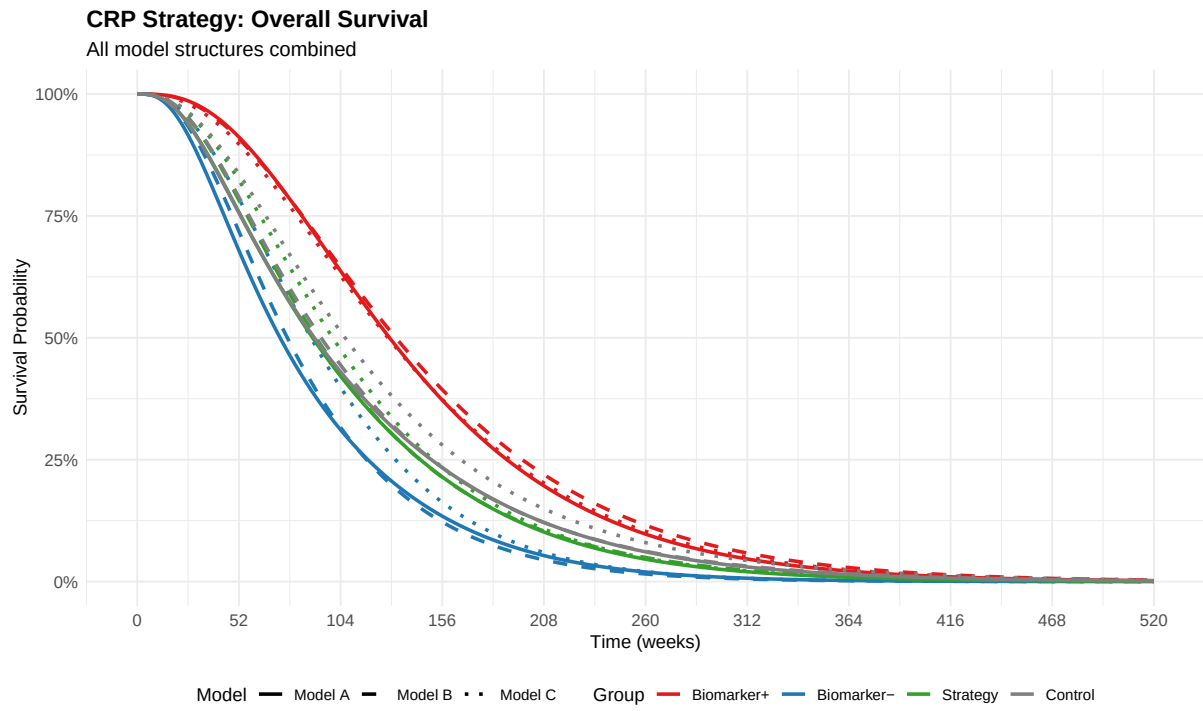


Figure 2: CRP strategy (prevalence: 33.8%): All model structures overlaid in a single panel for overall survival. Solid line = Model A, dashed = Model B, dotted = Model C.

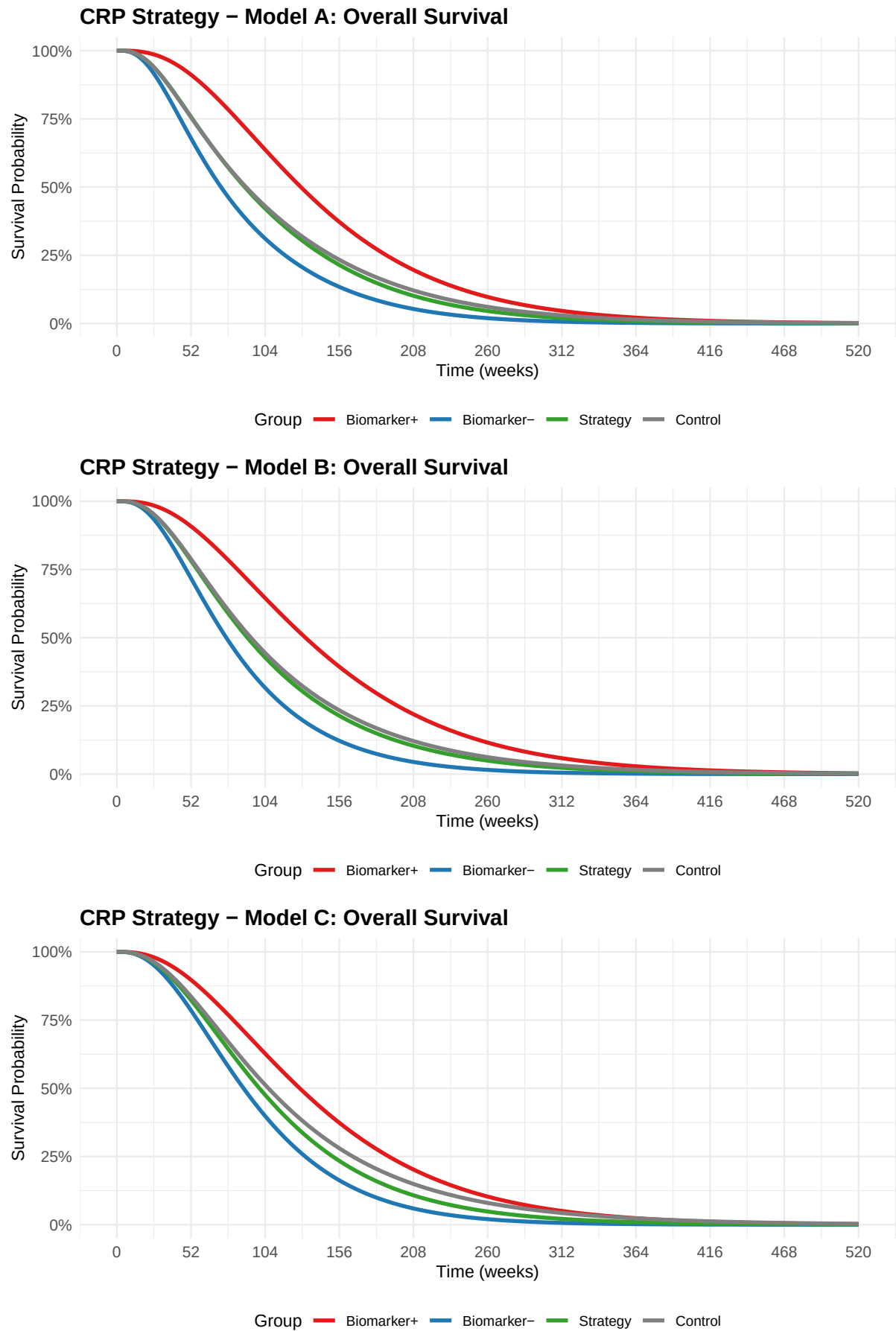


Figure 3: CRP strategy (prevalence: 33.8%): Individual model plots for overall survival showing detailed subgroup curves.

2.2 Progression-Free Survival

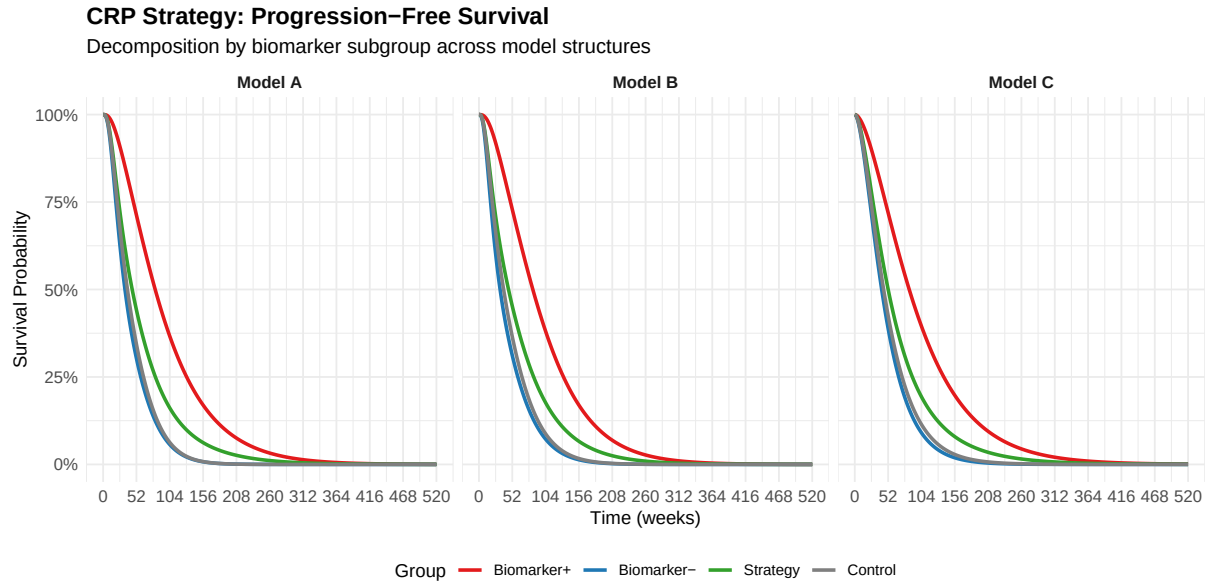


Figure 4: CRP strategy (prevalence: 33.8%) decomposition for progression-free survival across model structures.

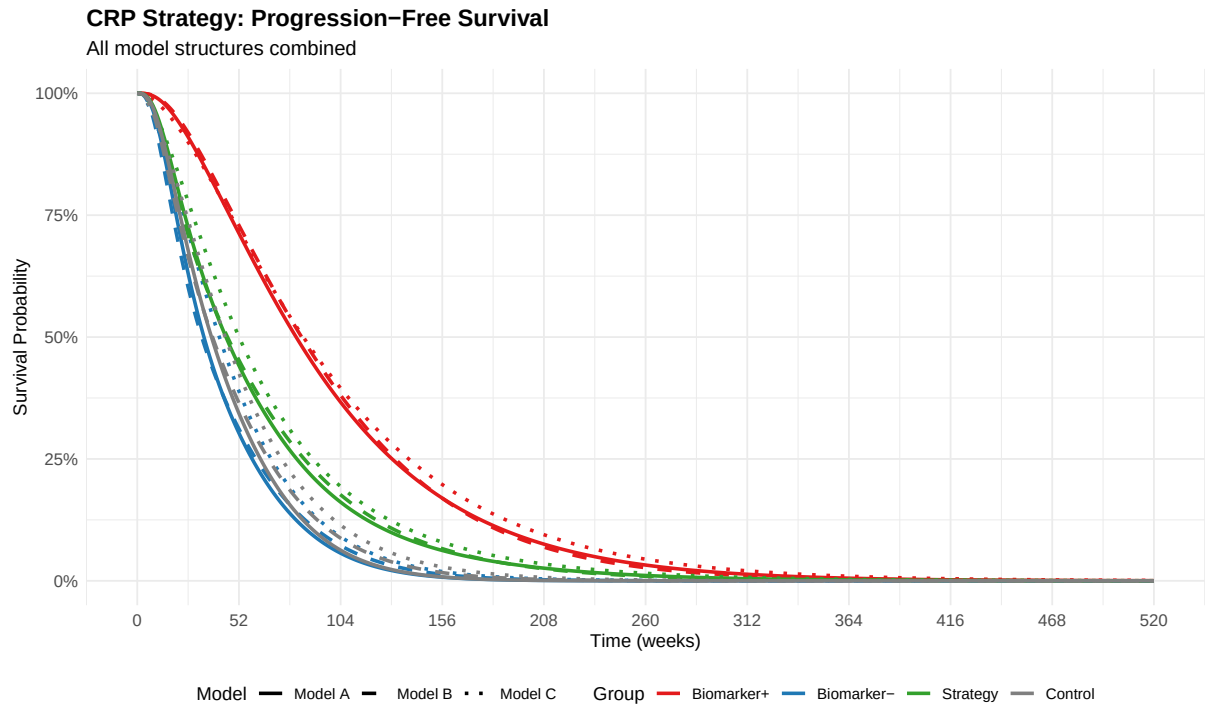


Figure 5: CRP strategy (prevalence: 33.8%): All model structures overlaid in a single panel for progression-free survival. Solid line = Model A, dashed = Model B, dotted = Model C.

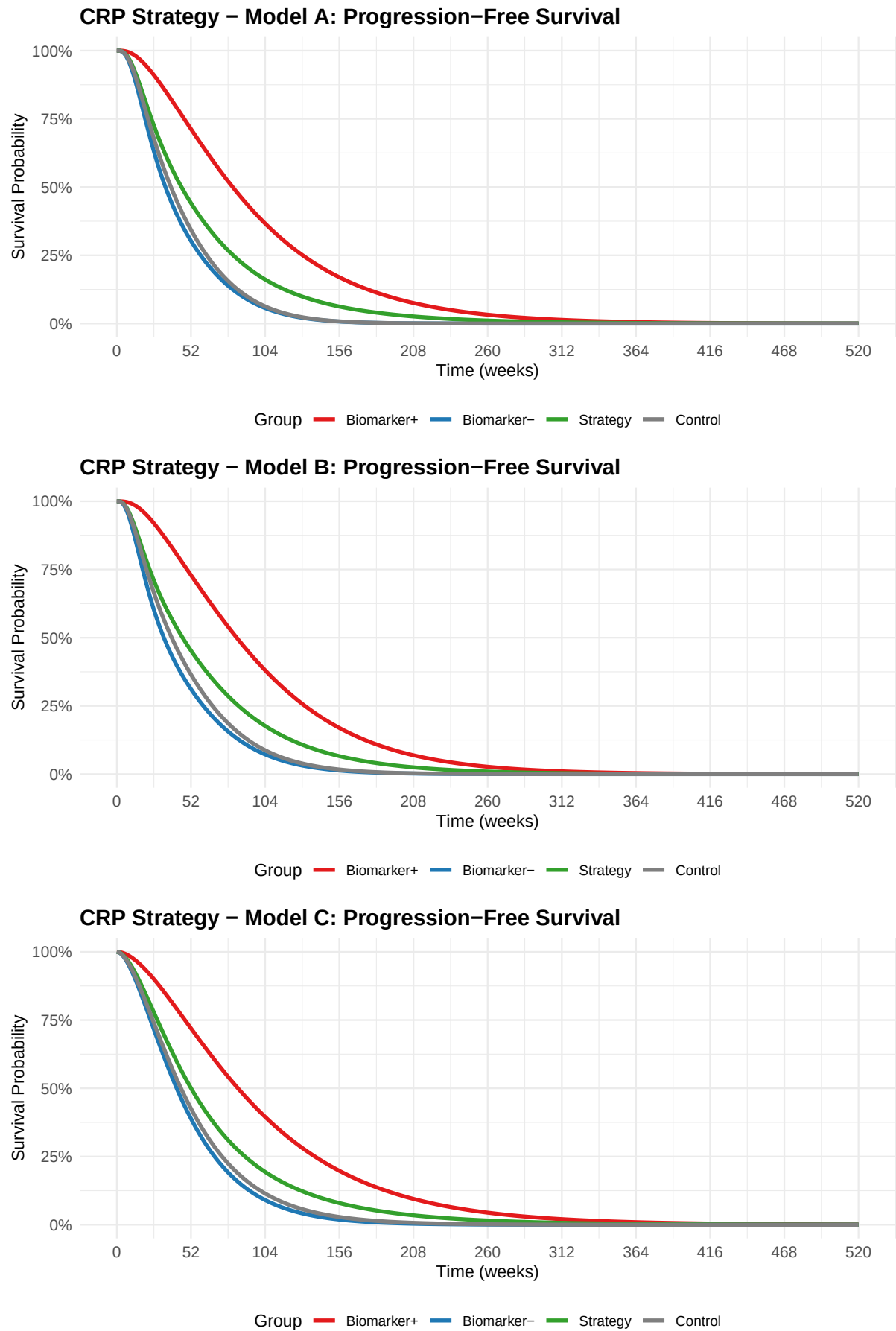


Figure 6: CRP strategy (prevalence: 33.8%): Individual model plots for progression-free survival showing detailed subgroup curves.

3 TLR Biomarker Strategy

Tumor lesion reduction (TLR) with a cutoff of $\geq 10\%$ for biomarker-positive status. **Biomarker prevalence: 64.7%**. This is an early response biomarker measured after initial treatment cycles. Biomarker-positive patients receive alternating FLOX and nivolumab; biomarker-negative patients receive FLOX only.

Table 4: TLR strategy: Overall survival life-years by model structure. Delta columns show difference vs. control.

Model	Control	Biomarker+	Delta+	Biomarker-	Delta-	Strategy	Delta Strategy
Model A	2.17	2.46	0.29	1.13	-1.04	1.99	-0.18
Model B	2.18	2.45	0.27	1.14	-1.05	1.99	-0.20
Model C	2.01	2.65	0.64	1.08	-0.93	2.10	0.09

Table 5: TLR strategy: Progression-free survival life-years by model structure. Delta columns show difference vs. control.

Model	Control	Biomarker+	Delta+	Biomarker-	Delta-	Strategy	Delta Strategy
Model A	0.90	1.60	0.70	0.45	-0.45	1.19	0.29
Model B	1.14	1.60	0.45	0.52	-0.62	1.22	0.07
Model C	0.96	1.89	0.93	0.49	-0.47	1.40	0.44

3.1 Overall Survival

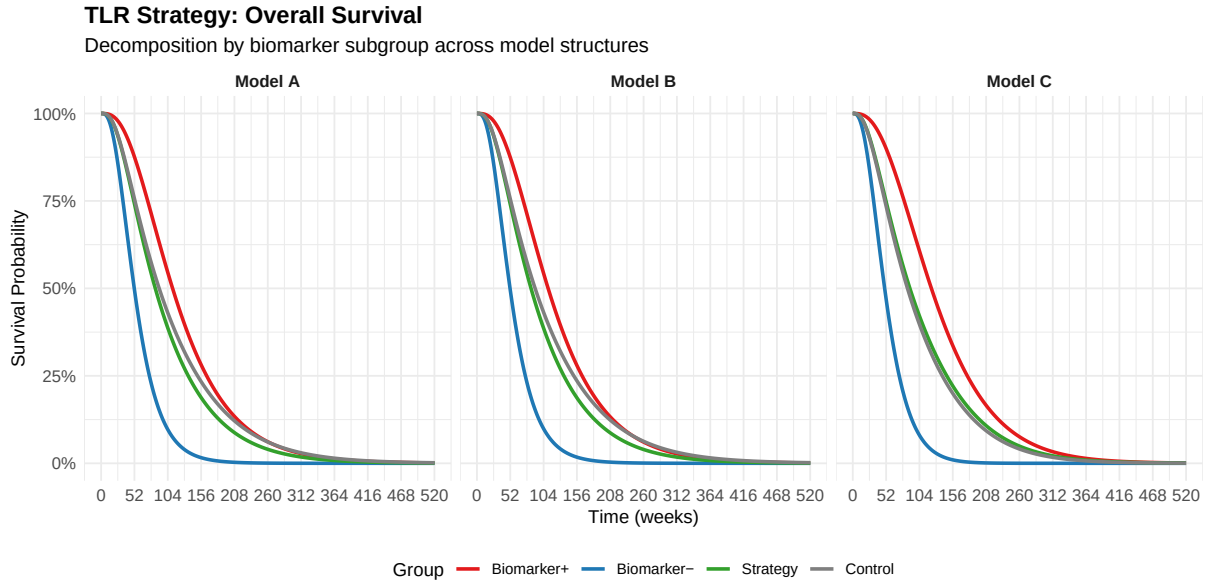


Figure 7: TLR strategy (prevalence: 64.7%) decomposition for overall survival across model structures. Red: biomarker-positive (experimental treatment), Blue: biomarker-negative (control treatment), Green: population-weighted strategy, Gray: control (all patients on standard treatment).

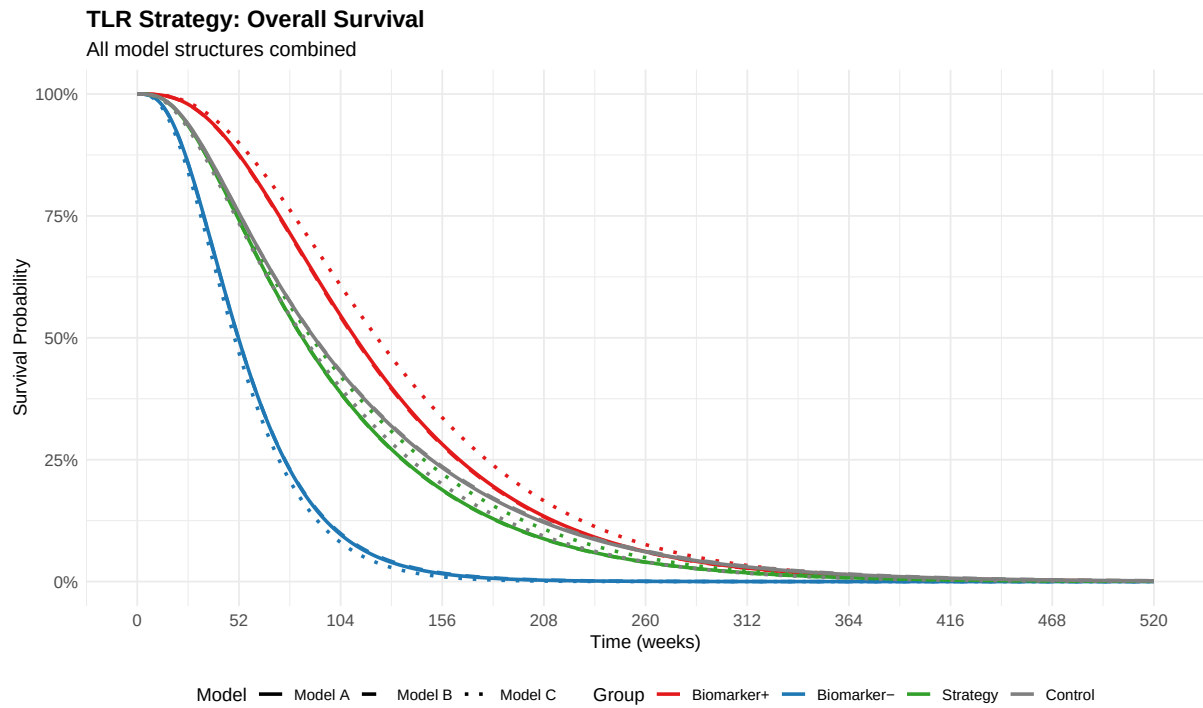


Figure 8: TLR strategy (prevalence: 64.7%): All model structures overlaid in a single panel for overall survival. Solid line = Model A, dashed = Model B, dotted = Model C.

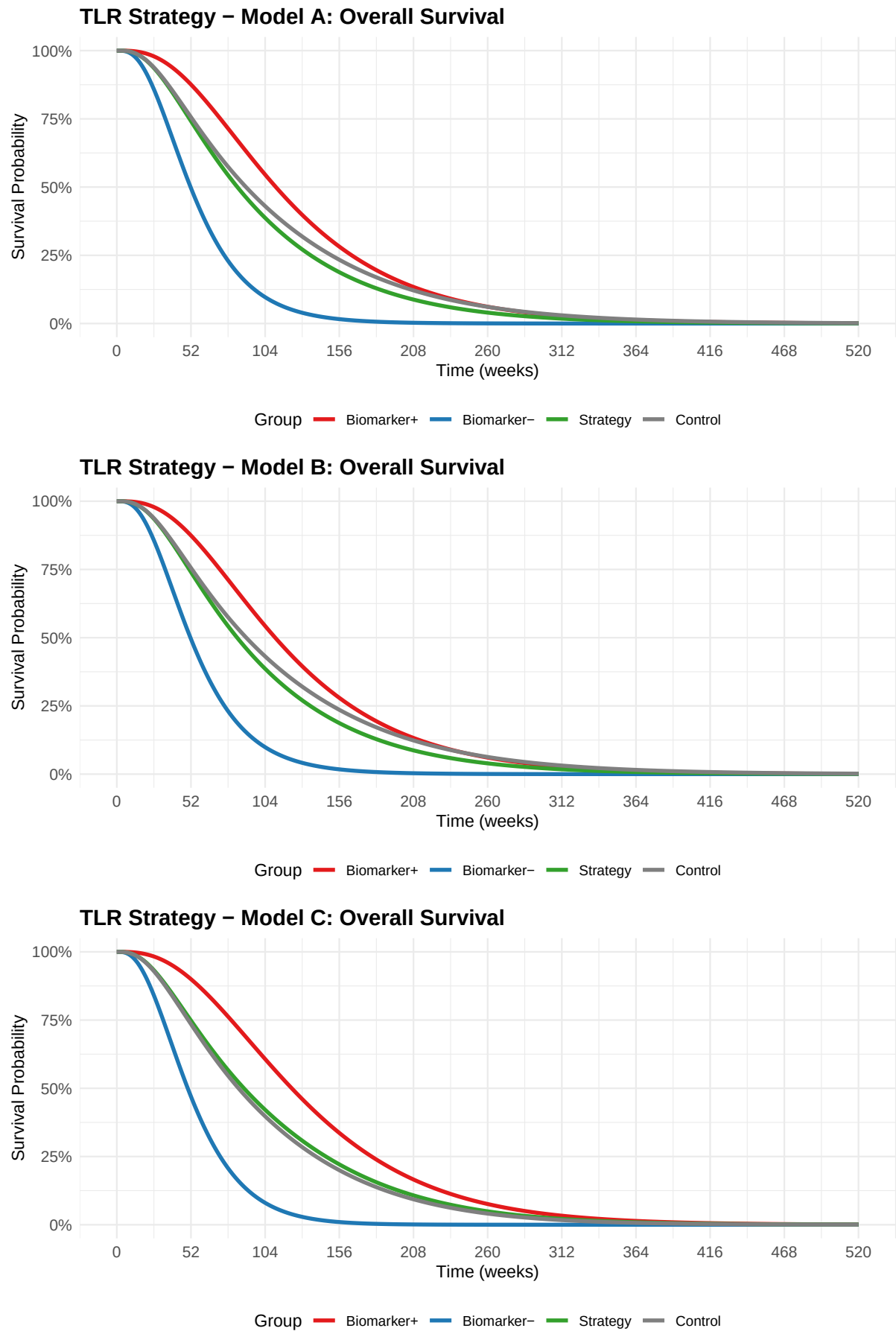


Figure 9: TLR strategy (prevalence: 64.7%): Individual model plots for overall survival showing detailed subgroup curves.

3.2 Progression-Free Survival

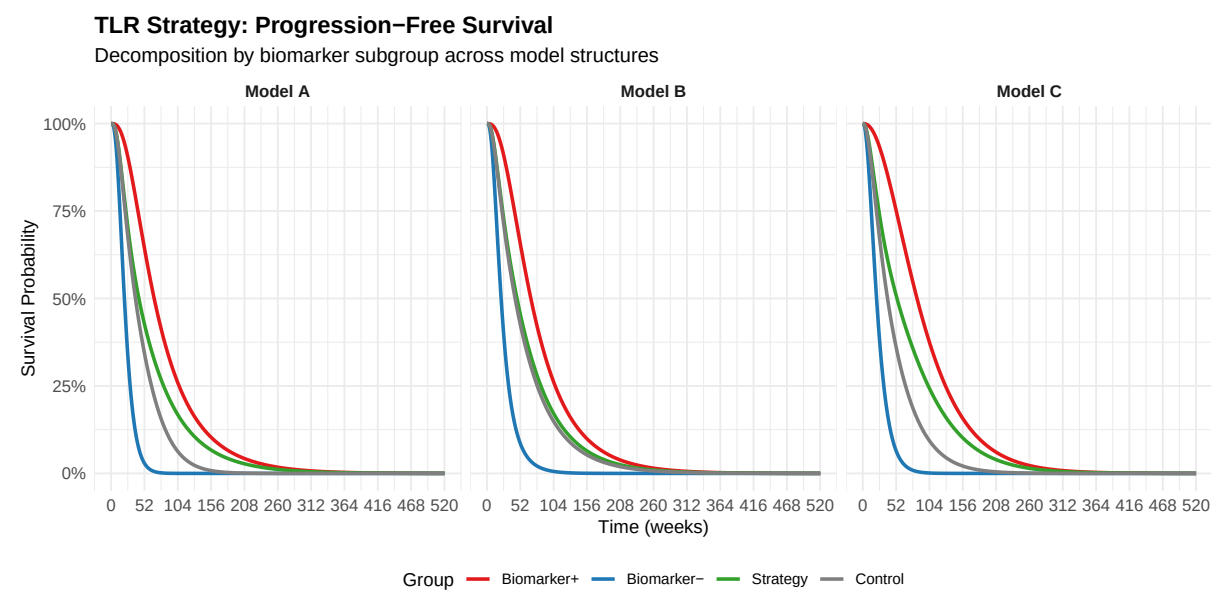


Figure 10: TLR strategy (prevalence: 64.7%) decomposition for progression-free survival across model structures.

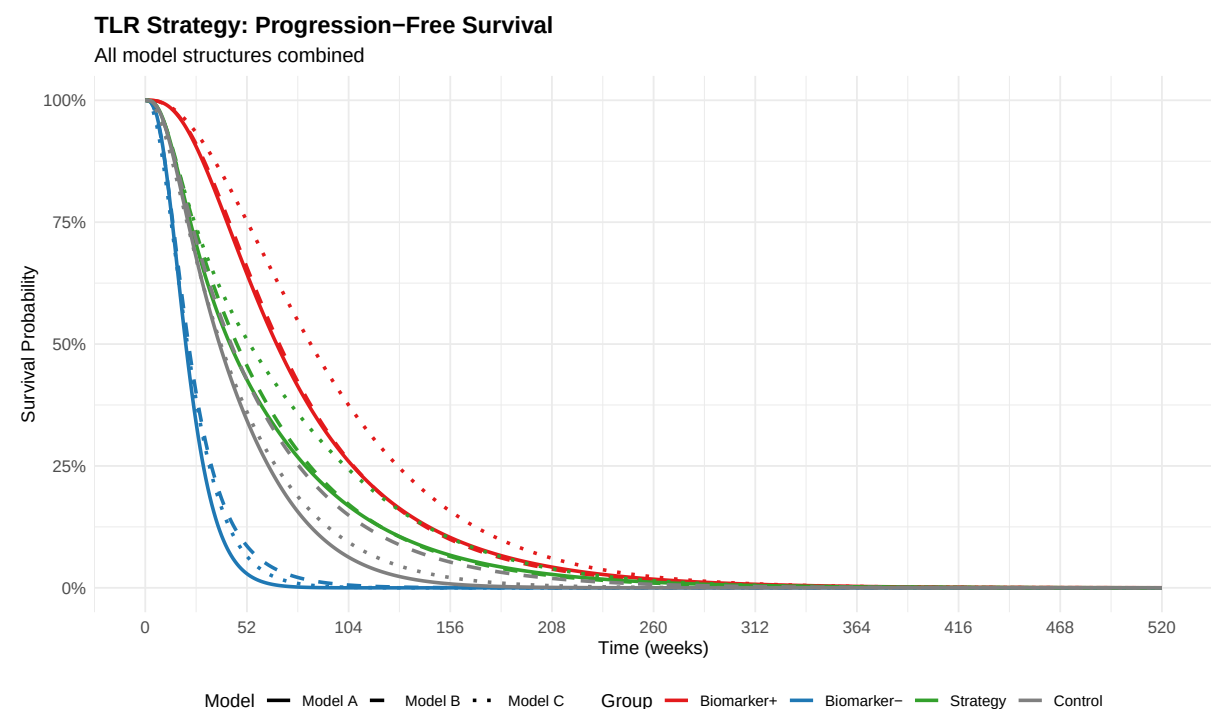


Figure 11: TLR strategy (prevalence: 64.7%): All model structures overlaid in a single panel for progression-free survival. Solid line = Model A, dashed = Model B, dotted = Model C.

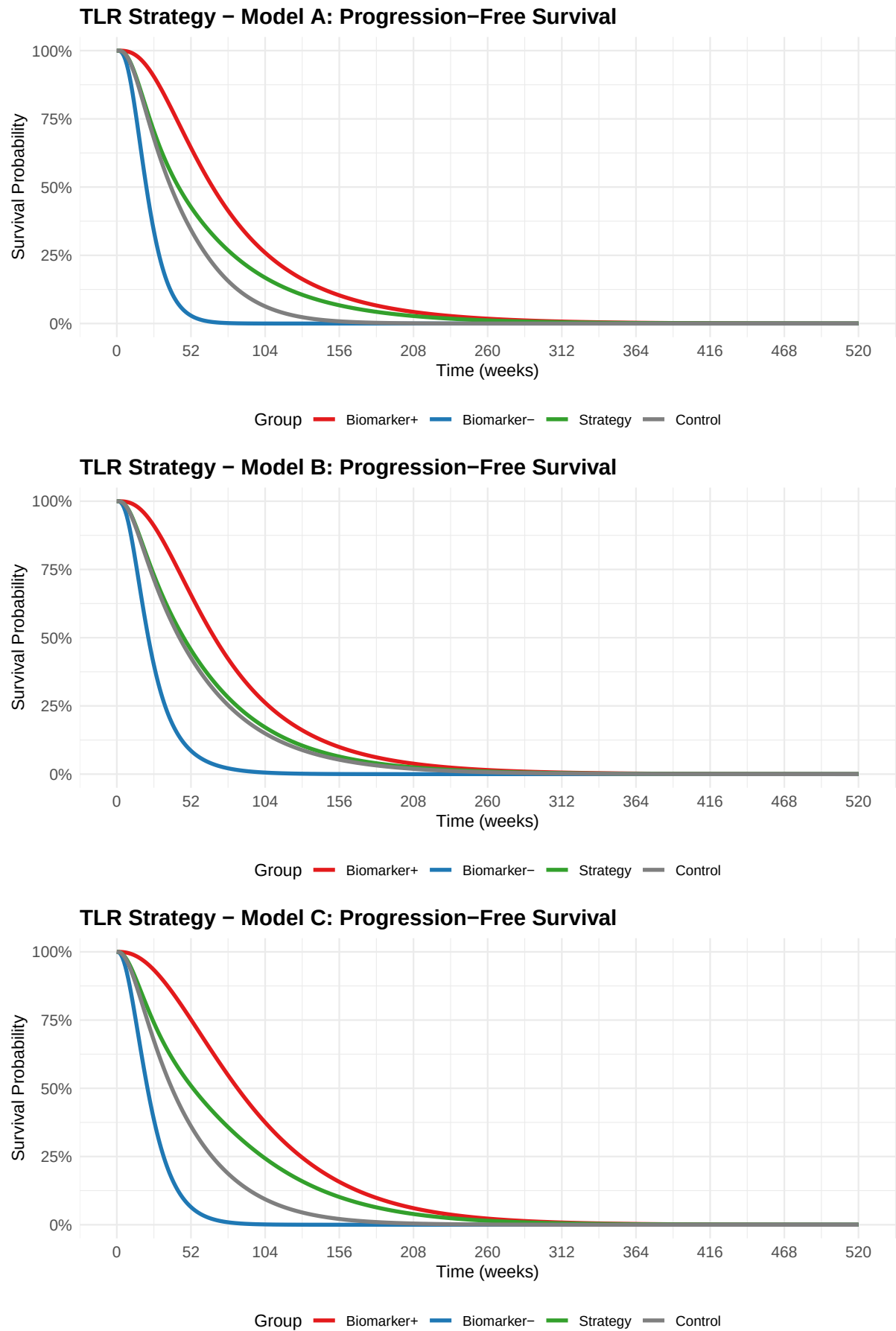


Figure 12: TLR strategy (prevalence: 64.7%): Individual model plots for progression-free survival showing detailed subgroup curves.

4 TMB/BRAF Biomarker Strategy

Combined genomic biomarker: either tumor mutational burden (TMB) ≥ 9 mutations/MB or BRAF V600 mutation positive (both determined by next-generation sequencing). **Biomarker prevalence: 44.9%**. Biomarker-positive patients receive alternating FLOX and nivolumab; biomarker-negative patients receive FLOX only.

Table 6: TMB/BRAF strategy: Overall survival life-years by model structure. Delta columns show difference vs. control.

Model	Control	Biomarker+	Delta+	Biomarker-	Delta-	Strategy	Delta Strategy
Model A	2.17	2.35	0.17	2.08	-0.10	2.20	0.02
Model B	2.14	2.43	0.29	2.07	-0.08	2.23	0.09
Model C	2.19	2.53	0.34	2.22	0.03	2.36	0.17

Table 7: TMB/BRAF strategy: Progression-free survival life-years by model structure. Delta columns show difference vs. control.

Model	Control	Biomarker+	Delta+	Biomarker-	Delta-	Strategy	Delta Strategy
Model A	0.90	1.49	0.59	0.86	-0.04	1.15	0.24
Model B	1.11	1.52	0.41	1.00	-0.10	1.24	0.13
Model C	1.04	1.96	0.92	0.96	-0.08	1.41	0.37

4.1 Overall Survival

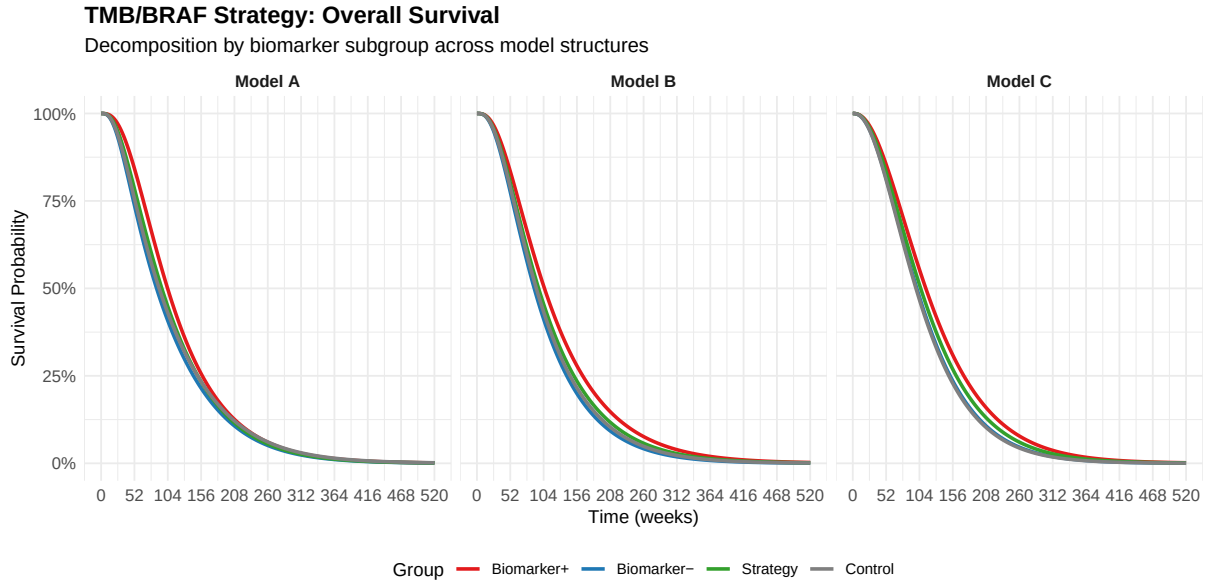


Figure 13: TMB/BRAF strategy (prevalence: 44.9%) decomposition for overall survival across model structures. Red: biomarker-positive (experimental treatment), Blue: biomarker-negative (control treatment), Green: population-weighted strategy, Gray: control (all patients on standard treatment).

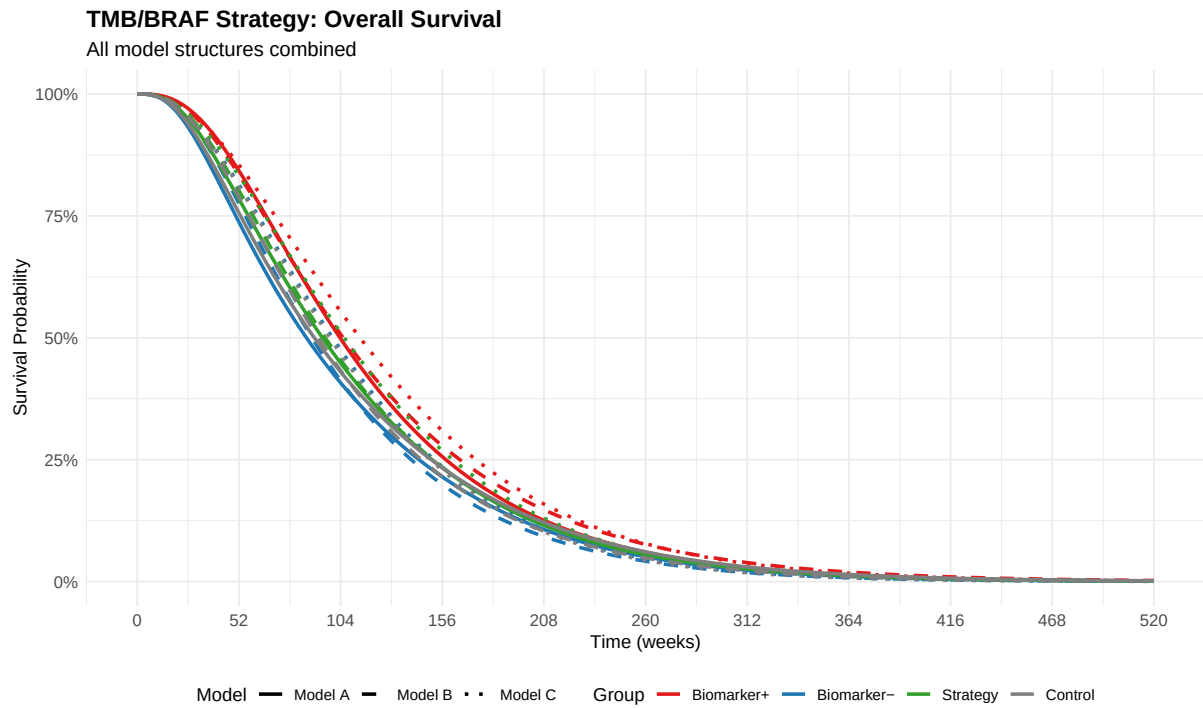


Figure 14: TMB/BRAF strategy (prevalence: 44.9%): All model structures overlaid in a single panel for overall survival. Solid line = Model A, dashed = Model B, dotted = Model C.

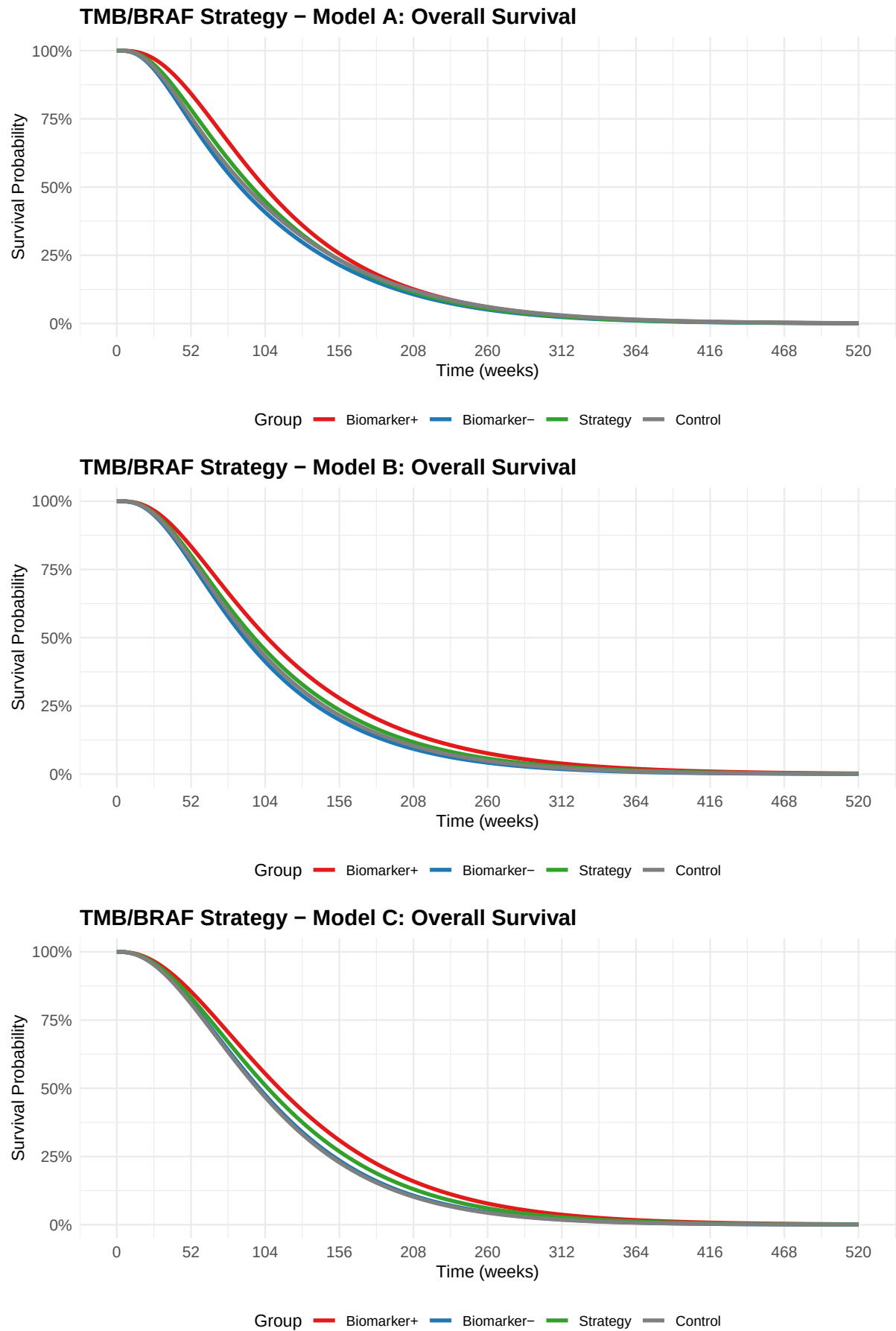


Figure 15: TMB/BRAF strategy (prevalence: 44.9%): Individual model plots for overall survival showing detailed subgroup curves.

4.2 Progression-Free Survival

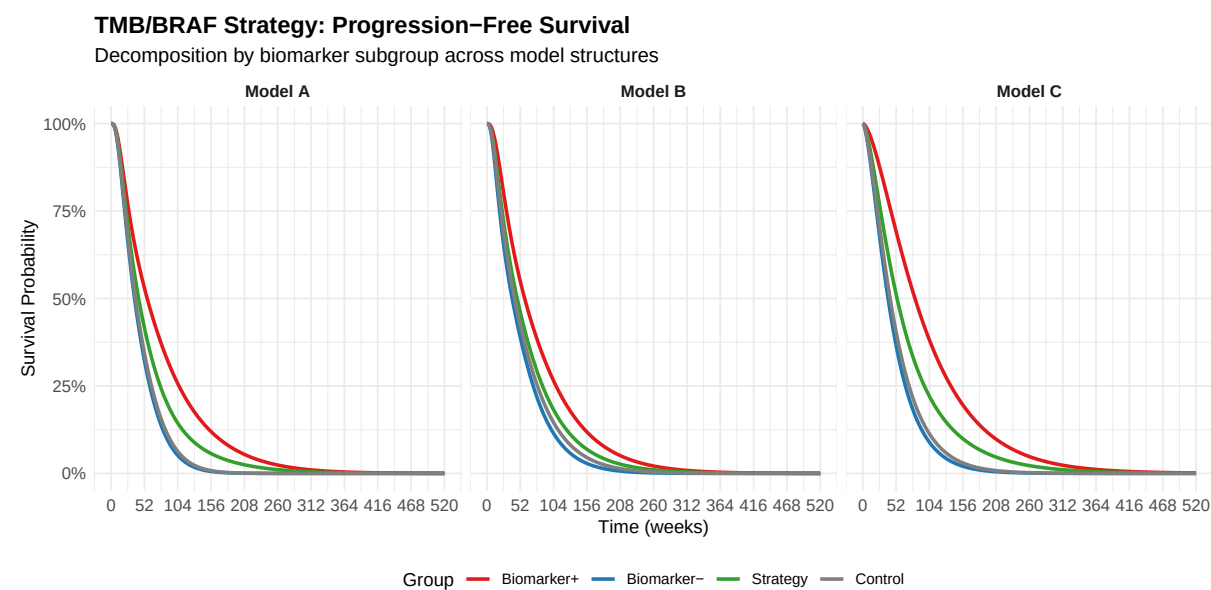


Figure 16: TMB/BRAF strategy (prevalence: 44.9%) decomposition for progression-free survival across model structures.

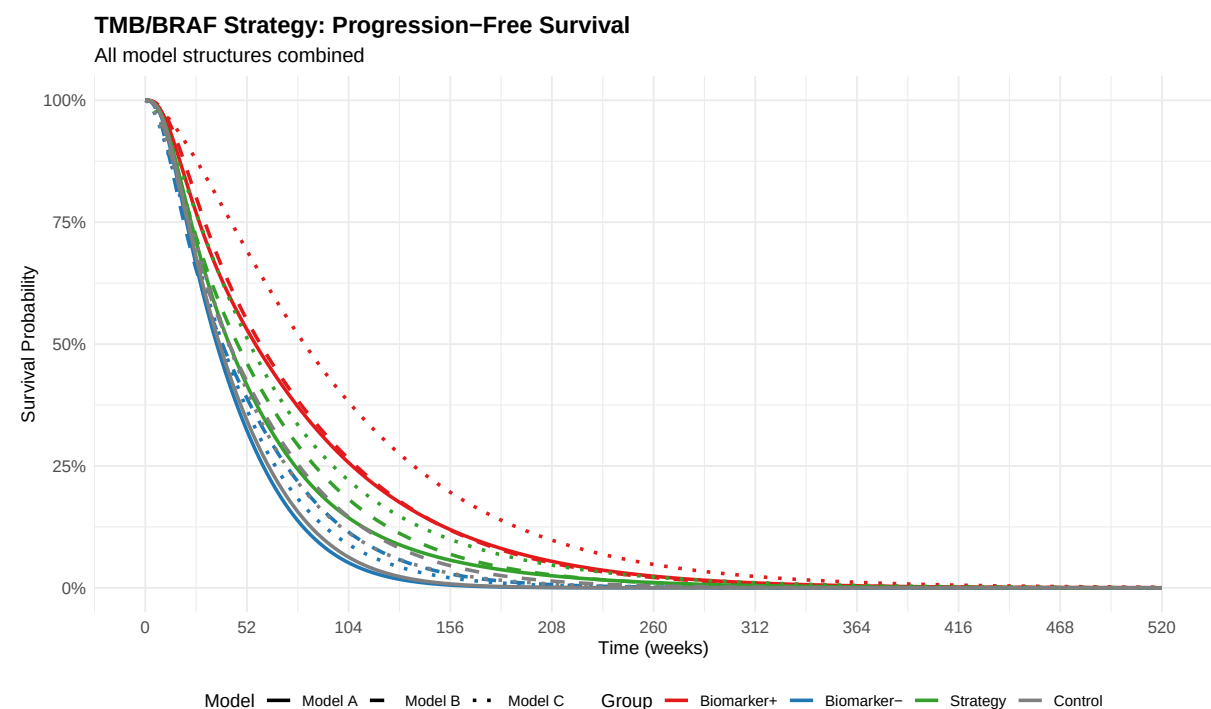


Figure 17: TMB/BRAF strategy (prevalence: 44.9%): All model structures overlaid in a single panel for progression-free survival. Solid line = Model A, dashed = Model B, dotted = Model C.

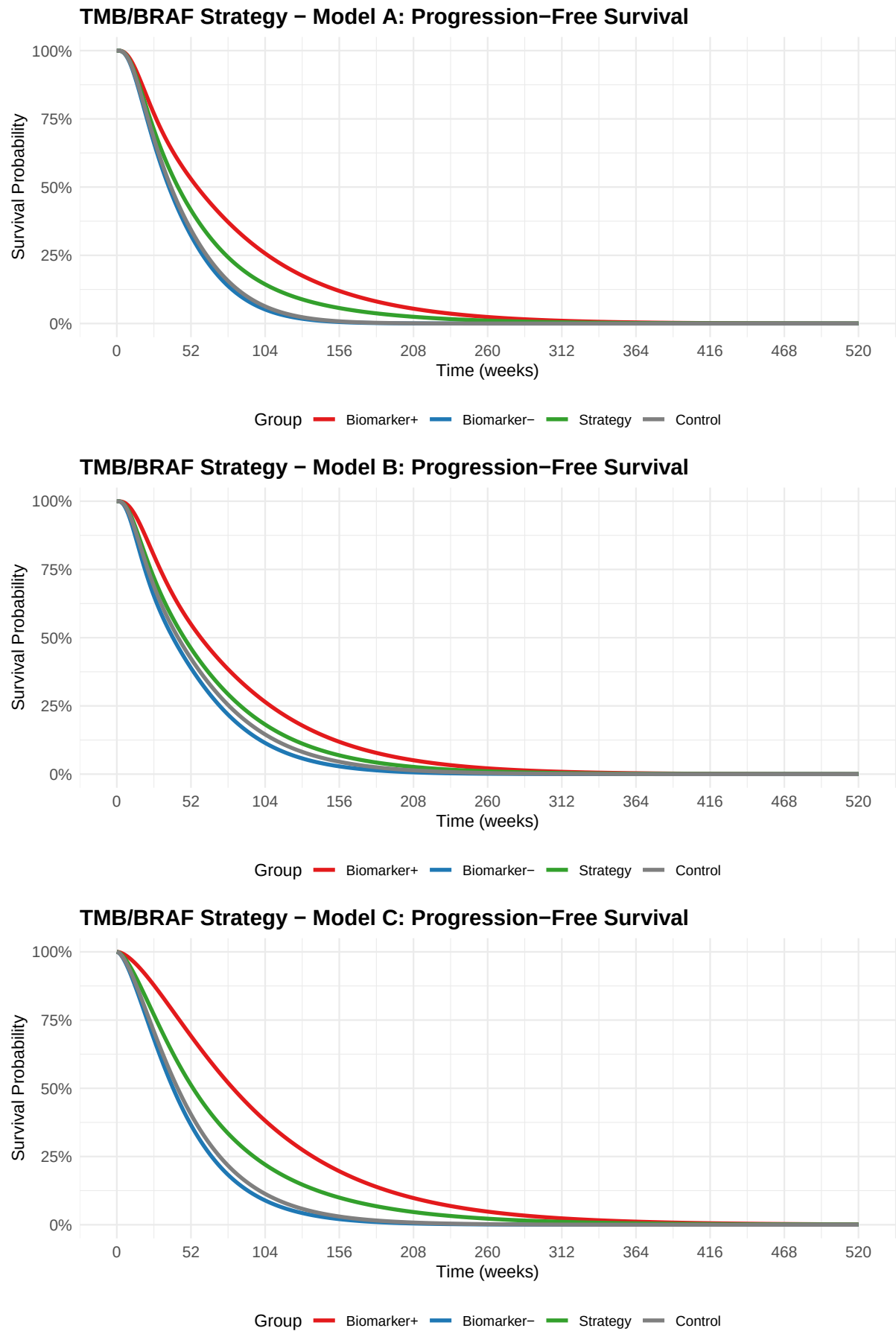


Figure 18: TMB/BRAF strategy (prevalence: 44.9%): Individual model plots for progression-free survival showing detailed subgroup curves.

5 Biomarker Prevalence Sensitivity

This section examines how strategy survival outcomes change with different biomarker prevalence rates. Using Model A (joint model with all biomarker interactions), we show the strategy survival curves at five prevalence levels (0%, 25%, 50%, 75%, 100%) for each biomarker.

At **0% prevalence**, no patients are biomarker-positive, so all receive control treatment - the strategy curve equals the biomarker-negative curve. At **100% prevalence**, all patients are biomarker-positive, so all receive experimental treatment - the strategy curve equals the biomarker-positive curve. Intermediate prevalence levels show the population-weighted combination of these two extremes.

5.1 Overall Survival

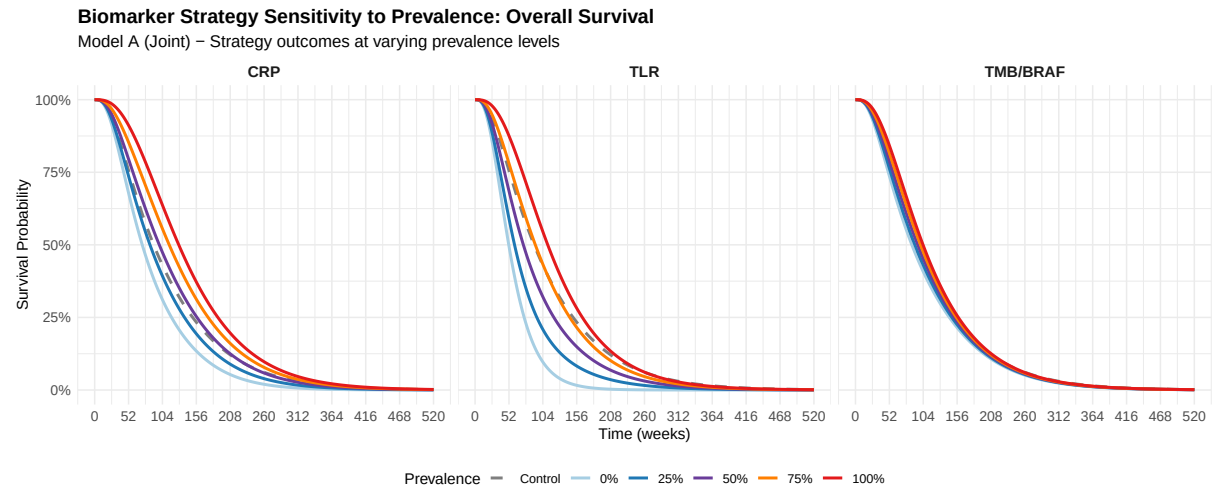


Figure 19: Strategy survival curves at varying biomarker prevalence levels using Model A (Joint). Control strategy (dashed gray) is unaffected by prevalence. Actual observed prevalences: CRP = 33.8%, TLR = 64.7%, TMB/BRAF = 44.9%.

5.2 Progression-Free Survival

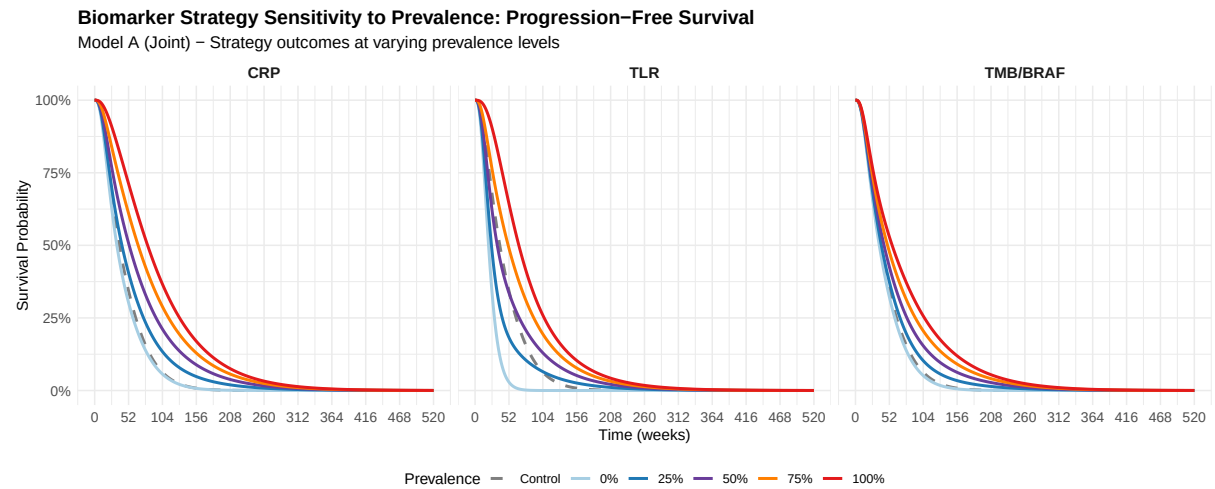


Figure 20: Strategy survival curves at varying biomarker prevalence levels using Model A (Joint). Control strategy (dashed gray) is unaffected by prevalence. Actual observed prevalences: CRP = 33.8%, TLR = 64.7%, TMB/BRAF = 44.9%.

6 Summary

All life-year (LY) gains reported in these tables are relative to the control strategy (standard of care with FLOX chemotherapy only). The tables show the decomposition of overall strategy effects into contributions from biomarker-positive and biomarker-negative patient subgroups.

Table 8: Summary of overall survival life-year gains relative to control, decomposed by subgroup.

	Biomarker	Model	Strategy	Biomarker+	Biomarker-
crp	CRP	Model A	-0.08	0.64	-0.45
crp1	CRP	Model B	-0.09	0.68	-0.48
crp2	CRP	Model C	-0.22	0.35	-0.51
tlr	TLR	Model A	-0.18	0.29	-1.04
tlr1	TLR	Model B	-0.20	0.27	-1.05
tlr2	TLR	Model C	0.09	0.64	-0.93
tmb_braf	TMB/BRAF	Model A	0.02	0.17	-0.10
tmb_braf1	TMB/BRAF	Model B	0.09	0.29	-0.08
tmb_braf2	TMB/BRAF	Model C	0.17	0.34	0.03

Table 9: Summary of progression-free survival life-year gains relative to control, decomposed by subgroup.

	Biomarker	Model	Strategy	Biomarker+	Biomarker-
crp	CRP	Model A	0.30	0.99	-0.06
crp1	CRP	Model B	0.27	0.96	-0.09
crp2	CRP	Model C	0.26	0.93	-0.08
tlr	TLR	Model A	0.29	0.70	-0.45
tlr1	TLR	Model B	0.07	0.45	-0.62
tlr2	TLR	Model C	0.44	0.93	-0.47
tmb_braf	TMB/BRAF	Model A	0.24	0.59	-0.04
tmb_braf1	TMB/BRAF	Model B	0.13	0.41	-0.10
tmb_braf2	TMB/BRAF	Model C	0.37	0.92	-0.08

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